COUNTRY PROFILES

Nuclear Power in Bulgaria

UPDATED TUESDAY, 17 FEBRUARY 2026

- Bulgaria has two nuclear reactors generating about 40% of its electricity.
- Bulgaria's first commercial nuclear power reactor began operating in 1974.
- Government commitment to the future of nuclear energy is strong.

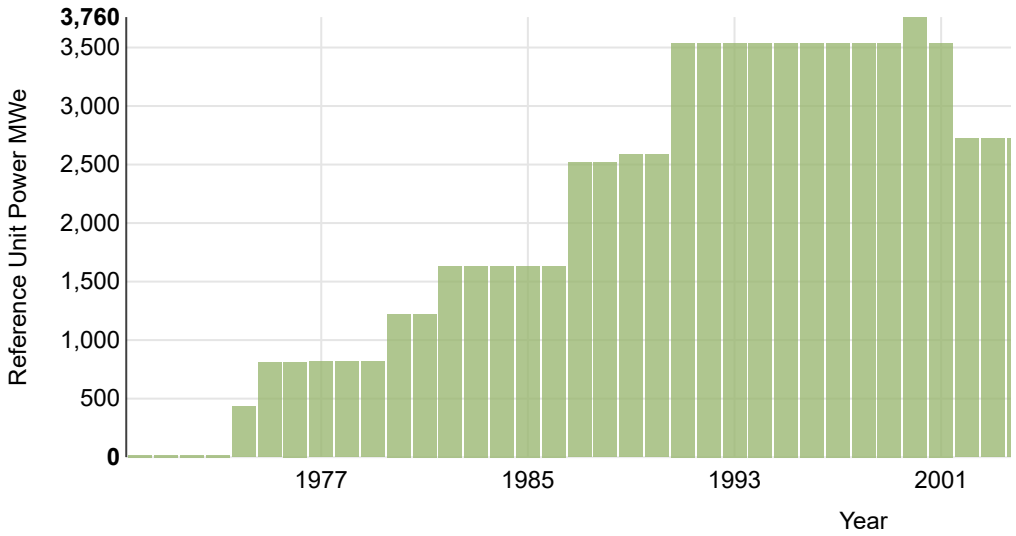
Privacy - Terms

OPERABLE REACTORS
2,006 MWe

REACTORS UNDER CONSTRUCTION
0 MWe

REACTORS SHUTDOWN
1,636 MWe

Operable nuclear power capacity



Electricity sector

Total generation (in 2023): 40.3 TWh

Generation mix: nuclear 16.2 TWh (40%); coal 11.7 TWh (29%); solar 3.5 TWh (9%); hydro 3.1 TWh (8%); natural gas 1.9 TWh (5%); biofuels & waste 1.9 TWh (5%); wind 1.6 TWh (4%); oil 0.3 TWh

Import/export balance: 3.3 TWh net export (4.4 TWh imports; 7.7 TWh exports)

Total consumption: 30.3 TWh

Per capita consumption: c. 4700 kWh in 2023

Source: International Energy Agency and The World Bank. Data for year 2023.

Electricity consumption in Bulgaria has been growing only slowly since 1980, but the country has in the past been a significant exporter of power[a].

A ten-year grid plan published in 2011 by Bulgaria's Electricity System Operator outlined investment of €390 million by 2015 to connect up to 1800 MWe of wind and 600 MWe of solar capacity, and over €300 million in the succeeding five-year period if the planned Belene nuclear plant were to be built. Forecasting very little growth in domestic demand (maximum 5400 MWe – 15% – to 2020), the plan made it clear that Belene's 15 billion kWh/yr would have been largely for export. In October 2023 plans to develop the Belene plant were cancelled. Plans are now proceeding for new capacity at Kozloduy.

Nuclear power industry

Reactors operating in Bulgaria

Reactor Name	Model	Reactor Type	Reference Unit Power (MWe)	Construction Start	First Grid Connection
Kozloduy 5	VVER V-320	PWR	1003	1980-07	1987-11
Kozloduy 6	VVER V-320	PWR	1003	1982-04	1991-08

Nuclear industry development

Since 1956 the Bulgarian government has favoured the use of nuclear power for electricity. The 2 MWt IRT-2000 research reactor started up in 1961 and in 1966 an agreement was signed with the Soviet Union for commercial units, providing the basis of the country's power programme. In the absence of Bulgarian safety

and regulatory bodies at that stage, these functions defaulted to Soviet standards.

The first pair of reactors were VVER-440 model V-230, built and commissioned in under five years. The second VVER-440 pair incorporated many of the much-improved safety features of the V-213 model[b]. The third pair were the larger VVER-1000 units, model V-320. All these were part of the Kozloduy plant, close to the Danube River border with Romania.

A second site was chosen near Belene, also near the Danube border with Romania. This was with a view to building four or six large units. Site works started in 1980 and construction of the first VVER-1000 V-320 unit started in 1987. This was partly built (40%, with 80% delivery of equipment), but was aborted in 1991 due to lack of funds.

Both NEK and Kozloduy Nuclear Power Plant plc are subsidiaries of Bulgaria Energy Holding (BEH). Kozloduy is the country's lowest-cost electricity producer.

In January 2023 the energy minister, Rosen Hristov, set out an energy strategy which includes plans for two new reactors at Kozloduy and two at Belene. The strategy outlines the continued use of coal until 2030 before reducing its use to zero by 2038. However in October 2023 the government formally cancelled the Belene project.

Kozloduy 1-4

During European Union (EU) accession negotiations with the European Commission (EC), Bulgaria committed to closing Kozloduy 1&2 by the end of 2002 and units 3&4 by the end of 2006[d]. All four units were VVER-440 model V-230 reactors, which the EC had earlier classified as non-upgradable. However, units 3&4 were of an improved design and closer to the later V-213 design than any others of their class[e]. Despite a 2005 opinion poll showing 75% support for keeping the two reactors running, the government finally ordered them to be shut down at the end of December 2006. Bulgaria joined the EU on 1 January 2007.

Kozloduy 5&6

An upgrade and modernization program for the V-320 units 5&6 extended to 2006, though the safety of these units was not in

question, as they conform well to international standards. Further plans to upgrade them by 2019 and extend operating lifetimes from 30 out to 60 years were implemented under a €360 million contract between Kozloduy Nuclear Power Plant and a consortium of Rosenergoatom and EDF from April 2012.

In October 2014 an agreement for refurbishment and lifetime extension of unit 5 was signed with Rosenergoatom, Rusatom Service and EDF. Rusatom Service said in May 2016 that technical work on unit 5 was complete to allow 30 years of further operation, and in October it provided documentation to enable the regulator, NRA, to extend the unit's licence eventually to 2047. It extended the licence by ten years initially.

A contract with a consortium of Rusatom Services and Bulgaria's Risk Engineering Ltd was signed in January 2016 to extend the operating licence of unit 6 to 60 years. The work took 30 months and a ten-year licence extension was granted in October 2019.

The government is strongly committed to the life extension and uprate to 104% of original capacity for both units. In 2013 a contract was signed with Rusatom Services to upgrade the turbine generator of unit 6, taking it to 1100 MWe by installing a new stator, with work completed in November 2015. A €24.7 million agreement with Rosatom in October 2015 was for upgrading the turbine generator of unit 5.

Following a vote in November 2022 to secure an alternative to Russian-supplied nuclear fuel, a month later, the government signed a 10-year deal with Westinghouse to supply fuel for Kozloduy 5. Later in December, an additional agreement relating to fuel supply for Kozloduy 6 during 2025-2034 was signed with Framatome. In May 2024 the first 43 Westinghouse RWFA (Robust Westinghouse Fuel Assembly) fuel assemblies were loaded into the Kozloduy 5 core, and in June 2025 the unit completed its first fuel cycle with the new fuel. A full transition from Russian fuel is expected by 2027.

Planned nuclear power reactors in Bulgaria

Reactor	Model	Type	Gross MWe
Kozloduy 7&8	AP1000	PWR	2x1250

KOZLODUY 7&8 TIMELINE

1980s

First proposals for units 7&8 at Kozloduy, when possible sites were reserved.

2010

A joint assessment by Kozloduy and Spain's Iberdrola reported that a seventh unit at the site would be viable, using the existing infrastructure. The government had been considering this, and a possible eighth unit, independently of plans for Belene.

2012

Westinghouse and GE had expressed interest in the project, using the AP1000 or a GE design respectively, but in April, the government said that the seventh unit would comprise what was to be Belene 1, an AES-92 plant supplied by Atomstroyexport. That month, the Council of Ministers approved in principle the construction of new capacity at Kozloduy. The Minister for Finance announced: "The government decided to open up for investment the project for a new reactor 7 at the Kozloduy nuclear power plant. It will be built on market principles, that is, without government money or state guarantees." A third-party investor is being sought for the project as licensing procedures get under way, and Kozloduy NPP-New Build plc was set up as the project company, a subsidiary of Kozloduy NPP plc, but with 49% available to a strategic investor following site licensing.

A request for proposals in June elicited responses from Areva, Areva/Mitsubishi, Bulgaria's Risk Engineering Ltd, Westinghouse and WorleyParsons.

In August the project company KNPP-New Build plc awarded a contract to Westinghouse Spain to assess the feasibility of two options: a VVER unit using Russian equipment already procured, but with instrumentation and control systems and fuel from Westinghouse, and a turbine-generator from Toshiba; and construction and operation of a western 1000-1200 MWe PWR, essentially Westinghouse's AP1000. The study was to evaluate the site, radioactive waste and used fuel management, use of existing infrastructure and facilities, licensing, local economic aspects, and the economics of the alternative reactor options.

Proposals for an environmental impact assessment were invited in September. A contract for site selection studies was awarded to Risk Engineering. In September Rosatom said that it would not cooperate with the Westinghouse Spain evaluation, and escalated its claim against NEK to €1 billion to encourage a purely Russian outcome for Kozloduy 7. See also following section on The Belene saga.

2013

In May the results of studies and assessments were to be presented to the Council of Ministers for decisions on the technology, the number of units, the capacity, and construction schedule. From November the Economy and Energy Minister was in negotiations with Westinghouse, and said that construction of an AP1000 reactor as unit 7 might begin in 2016. Evidently the Bulgarian Nuclear Regulatory Agency (NRA) will accept US design certification for the AP1000 with one year of additional review.

2014

Westinghouse in mid-2014 signed agreements with three local companies for major aspects of the project. Westinghouse would be responsible for providing all of the plant equipment,

design, engineering and fuel for the new unit.

Bulgarian Energy Holding (BEH) progressed these negotiations including the prospect of Toshiba investing in the project to the extent of a 30% share. The Turkish Exim Bank has expressed readiness to finance the project. Bulgaria was hoping that a Toshiba stake would allow access to funding from the US Export-Import Bank and the Japan Bank for International Cooperation. However, in June Toshiba withdrew from negotiations regarding investment, and handed over to Westinghouse which on 1 August signed an agreement to take 30% equity in the new plant through Kozloduy NPP-New Build plc. Government-owned Kozloduy NPP plc will hold 70%, and finance would need to be secured by both parties. The agreement also formalized the selection of an AP1000 design reactor by Bulgarian Energy Holding (BEH), its subsidiary Kozloduy NPP plc and Kozloduy NPP-New Build plc. Westinghouse quoted $7.7 billion for the plant, with estimated completion in 2025. At the time, Kozloduy NPP-New Build expected regulatory siting approval for units 7&8 early in 2017.

Bulgaria's Council of Ministers approved an Economy and Energy Ministry report on the shareholder agreement on 30 July. The agreement – including the financing terms of an engineering, procurement and construction (EPC) contract for the project – were to enter into force after approval by the new government, following elections in October. Negotiations since November 2013 had involved all political parties.

2015

Progress was delayed by lack of finance and low electricity demand. In April Westinghouse announced that while the shareholder agreement for Kozloduy 7 had expired, discussions were continuing with Kozloduy NPP on a new structure and timeline. "While there is unanimity that the project is clearly attractive in the long-term, the parties believe that different models will need to be considered for deploying the AP1000 technology in the future," according to Westinghouse.

2016

Industry association Bulatom said that the government would require a partner to provide the project and financing in one package, without state guaranteed long-time prices for the electricity produced. Also there will be no integration with the existing units (5&6) including the prices of the electricity produced by them. In December the government said it could not afford to proceed, and having considered the prospect of taking a 30% share, Westinghouse did not want to invest in the project itself.

2020

In February the NRA gave site approval on the basis of an environmental impact assessment which considered AP1000, AES-92 and AES-2006 plants. The energy minister said that it should be a different technology to Belene, hence AP1000.

In October the prime minister said that the country wanted to build unit 7 with "...completely different technology, with different nuclear fuel," and confirmed that they will consider US nuclear technology. Later in October Bulgaria and the USA signed a Memorandum of Understanding concerning strategic civil nuclear cooperation.

2021

In January the Bulgarian government approved plans for a new unit at Kozloduy.

2023

In January, Bulgaria's National Assembly voted by 112 to 45, with 39 abstentions, in favour of a draft decision asking ministers to negotiate with the US government for the new AP1000 unit at Kozloduy and urged action by 1 March to speed up the process for approval and construction of the unit as well as initiating a licensing and environmental impact assessment procedure for another reactor, which would be unit 8 at Kozloduy.

In October the Bulgarian Council of Ministers gave the go-ahead for the construction of unit 7 and the beginning of preparatory work for unit 8 of the Kozloduy plant. The government set a target date for the first unit to be ready of 2033, with funding of $270 million provided to the project company. The second unit is to be completed two or three years after the first.

2024

In March the parliament ratified an agreement with the USA on cooperation regarding the construction of Westinghouse AP1000s for Kozloduy 7&8.

In November an engineering services contract was signed with Westinghouse and Hyundai E&C for two AP1000 reactors, with a target for unit 7 to be operational in 2035 and unit 8 in 2037.

2025

In February a site permit application was submitted to the Nuclear Regulatory Agency.

In July 2025 the Ministry of Energy agreed a partnership with Citi bank. Citi would act as exclusive coordinator and lead arranger of export credit.

In December 2025 a three-company consortium of Laurentis, BWXT and CNPSA signed an owners' engineer contract to advance the two planned AP1000 units at Kozloduy.

The Belene saga timeline

1980s & 90s

Site works at Belene near the Danube border with Romania started in 1980 and construction of the first VVER-1000 V-320 unit started in 1987. This was partly built (40%, with 80% delivery of equipment) but was aborted in 1991 due to lack of funds.

2003

Five reactor vendors expressed interest to the Energy Ministry in completing Belene or building new units there, after the Bulgarian government considered completing the Belene plant in the run-up to the closure of the first two units of Kozloduy.

2005

The government approved the construction of Belene as a 2000 MWe plant. Parsons E&C Europe (now WorleyParsons) was appointed architect-engineer for the project to oversee redesigning it. Two consortia submitted bids to build the plant. Both had two variants: using the old VVER-1000 V-320 equipment already on site, and building afresh two AES-92 units.

2006

In October the National Electricity Company (*Natsionalna Elektricheska Kompania*, NEK) chose Atomstroyexport (ASE) over a Skoda-led consortium to build the plant comprising two 1060 MWe capacity third-generation VVER-1000 (AES-92) units[f]. Russia's ASE led a consortium including Areva NP, Siemens and Bulgarian enterprises in the project. The contract was signed in January 2008 for €4 billion excluding first cores but with allowance for inflation adjustment.

2007

The European Commission responded positively to the choice of reactor technology, saying: "Compared to previous VVER-1000 Russian reactor types in operation, the AES-92 should comprise advanced technology passive safety systems and are considered to be in compliance with European Utility Requirements (EUR)."[1]

2008

Construction at Belene was officially restarted in September – though only preliminary site activities were carried out – and was then suspended until a strategic investor for the project could be found. Completion of the units had originally been expected in December 2013 (unit 1) and June 2014 (unit 2).

Meanwhile, in December ASE signed an €82 million contract with Russia's OMZ Izhorskiye Zavody to supply reactor pressure vessels, steam generators and related equipment, total 1300 tonnes (The €82 million contract figure is probably net of the V-320 equipment returned to ASE for use by Atomenergoprom at Kalinin in Russia). The first pressure vessel was scheduled to be shipped in December 2011 and the second in September 2012.

2009

In June a RUR 2.8 billion (€65 million) contract was signed for the steam generator shells to be finished by ZiO-Podolsk and shipped 2010-12.

In April ASE and Areva NP signed a contract to develop documentation on the instrumentation and control (I&C) systems.

In September the new government of the Citizens for European Development of Bulgaria (GERB) party (which came into office in July) said that it would not take a 51% stake in the Belene project as had been planned, and that 20% looked more practical, to be held by Bulgarian Energy Holding (BEH). German utility RWE had been set to take a 49% stake in the project (with NEK holding the remaining 51%), but pulled out in October[g].

2010

Russia offered a loan to finance construction activities until a strategic investor was found, but this offer was rejected by the Bulgarian government[h]. In November NEK signed a memorandum of understanding (MoU) with Rosatom to establish a new project company – Belene Power Company – with initially 51% being held by NEK. For its part, Rosatom was to endeavour to arrange financing for the project, attract other investors, and facilitate ASE achieving commissioning of the reactors by 2016 and 2017.[2] NEK would transfer the site and work to date as in-kind contributions. Development of the plant's economic model and the selection of investors was to be carried out by HSBC, which was selected by BEH as project consultant in November. The non-binding agreement called for Belene Power Company to be

established before April 2011 and for construction to resume before October 2011. However, there remained several issues to be resolved, notably the cost escalation provisions for the project[i].

2011

In March the Energy & Economy minister said that the equipment being built for Belene might be installed alongside similar units at Kozloduy instead, though it was not clear how that would resolve the financing question. In July Atomstroyexport took NEK to the international court for non-payment of $58 million for equipment supplied. However, it said that that the case "should in no way be construed as a measure to pressure the customer in order to accelerate decision-making" and that it respected NEK's desire to carefully consider the Belene project. High-level negotiations continued on implementing the project, particularly its financial structuring.

In September AtomStroyExport and NEK signed a supplement to their agreement on the construction of the Belene plant, extending it until the end of March 2012. While NEK stalled as it sought some $10.5 billion of Western investment funding, ASE said that work on the foundation pit for the first reactor at Belene had been completed and other preparatory infrastructure was in place.

2012

In March the government said that it would pay for one set of VVER-1000 hardware which has already been manufactured and install it at Kozloduy as an AES-92 unit. This apparently amounted to about €500 million. However, Rosatom claimed a further €500 million in damages for the aborted project, and said that it might accept the assets of NEK as settlement.

2013

In February after the resignation of the prime minister and cabinet, parliament confirmed that the project would be abandoned in favour of building a new unit at Kozloduy. Then, late in 2014 following an election, the matter was again under consideration.

Rosatom was keen to supply fuel for 60 years, and said that work to the value of €2.5 billion could be available for Bulgarian companies, mostly in engineering.

ASE took the matter to the International Court of Arbitration in Geneva, claiming a total of €1.2 billion.

2016

In June the court ruled in favour of ASE. At the end of the year, NEK agreed to pay ASE €620 million in compensation for the already-produced primary cycle equipment for Belene units 1&2. The amount was settled by year end, and ASE agreed to drop its claim for €23.8 million interest.

2018

In May the prime minister announced that the government would ask parliament to revive the project, after China National Nuclear Corporation (CNNC) said it was interested in building the plant. In June six years after the Belene project was last put on hold, parliament voted to resume the search for investors.

2019
In May the government advertized for a strategic investor to participate in the Belene project to build two large reactors, but it said that neither funding guarantees nor long-term electricity sales contracts would be offered. In August the energy minister reported that 13 applications for participation had been received, including seven to be a strategic advisor, four to be an electricity customer and/or minority shareholder, and two to be an equipment supplier. In December a shortlist of five companies was announced: CNNC, Rosatom and Korea Hydro & Nuclear Power, as well as Framatome and General Electric (GE) for equipment contracts.
2020
In June Rosatom, Framatome and GE agreed in principle to form a consortium to bid for the project.
2021
In January the government approved plans for a new reactor Kozloduy, following discussions with Westinghouse about making maximum use of Russian equipment purchased for the suspended Belene plant.
2023
In October the Bulgarian government announced that it had reversed the decision to develop the Belene plant.
2025
In March Ukraine agreed to purchase equipment from the unfinished Belene plant to be used at Khmelnitsky 3&4. However, in April the Bulgarian government announced that it would stop the sale, citing interest from American companies in nuclear-powered data centres.

Small reactors

In October 2021 Bulgarian Energy Holding (BEH) – which owns one of the country's five coal-fired power stations – signed an agreement with Fluor to examine the feasibility of replacing those plants with 77 MWe NuScale reactors in modules of four, six or 12 units. In December 2024 the US Trade and Development Agency signed a grant with BEH for a technical analysis of US-sourced SMR design options and site evaluation.

Fuel cycle

All front end fuel cycle services in Bulgaria were historically provided by Russia's TVEL through Techsnabexport (Tenex). As part of a programme to diversify its nuclear fuel supply, Kozloduy

Nuclear Power Plant signed fuel supply contracts with Westinghouse for unit 5 and Framatome for unit 6 in December 2022. The first Westinghouse fuel assemblies were loaded into unit 5 in May 2024.

Radioactive waste management

The State Enterprise Radioactive Waste (SE-RAW) is responsible for waste management.

Under a 2002 agreement, Bulgaria was paying Russia $620,000 per tonne of used nuclear fuel repatriated for reprocessing in the Mayak plant at Ozersk, though some was also been sent to the Zheleznogorsk plant at Krasnoyarsk[i]. Shipments were halted in 2022 following Russia's invasion of Ukraine.

Used fuel is initially stored in pools at each reactor, but in 1990 a pool-type storage facility was constructed at Kozloduy to take fuel from all the units. This was upgraded and a new licence issued by the Bulgarian Nuclear Regulatory Agency (NRA) in 2001.

A new €49 million dry storage facility for 2800 VVER-440 used fuel assemblies has been built near this at Kozloduy, with finance from the Kozloduy International Decommissioning Support Fund administered by the European Bank for Reconstruction and Development (EBRD). This Dry Spent Fuel Storage Facility (DSFSF) was being constructed by a joint venture partnership between Nukem Technologies and GNS. Later expansion to accommodate 8000 VVER-440 and 2500 VVER-1000 assemblies is envisaged. The facility, with capacity of 5200 fuel assemblies in 72 casks, was officially opened in May 2011. It will accommodate used fuel from Kozloduy's four closed VVER-440 units, currently in pool storage, and will be subsequently enlarged to receive casks with fuel from VVER-1000 units 5&6.

Also at Kozloduy is a low- and intermediate-level radioactive waste treatment and storage facility.

In mid-2005 the Council of Ministers resolved that a 50,000 cubic metre national near-surface low-level and intermediate-level waste (LILW) disposal facility should be constructed by SE-RAW for operation in 2015. The National Repository for Disposal of Radioactive Waste (NRD RAW) will accept wastes over the next 60 years and store it for some 300 years in a near-surface, trench-type repository. The €120 million project will be paid for from the radioactive waste fund of the EBRD. It will be built in successive

modules to reach an eventual capacity of 138,200 cubic metres (345,000 tonnes) of wastes. (For comparison, in 2010 the Kozloduy plant generated some 1400 m^3 of LLW and ILW.) In April 2009, SE-RAW awarded a €2.6 million three-year contract to a consortium comprising of Empresarios Agrupados Internacinal SA, VT Nuclear Services and ENPRO Consult of Bulgaria for project management. In October 2011 SE-RAW awarded a contract to Westinghouse Spain, Enresa (Spain) and DBE Technology from Germany to design the repository. In July 2013 it was announced that a 45 ha site adjacent to Kozloduy had been selected for it. In July 2016 SE-RAW signed a €72 million agreement with a consortium led by Nukem with four Bulgarian companies to build the first phase of the facility to accept Kozloduy decommissioning wastes. The project is funded by a grant from the Kozloduy International Decommissioning Support Fund, administered by the EBRD.

A near-surface repository, Novi Han, about 35km southeast of Sofia, was licensed for non-nuclear radioactive waste. In January 2026 the NRA issued a decommissioning licence for the facility, with waste to be transferred to the new repository at Kozloduy once it is operational.

Two national funds were created in 1992, one for the safe disposal of radioactive waste and one for the decommissioning of nuclear facilities, but these funds did not function properly until 1999. The funds are independent from the nuclear industry and managed by the government. The main contribution to the two funds comes from an electricity price levy specified by the Bulgarian Council of Ministers. The Kozloduy nuclear plant pays 3% of the price of its power into the waste management fund and a further 7.5% into the decommissioning fund.

Reactor decommissioning

In addition to the national decommissioning fund, in June 2001 the Kozloduy International Decommissioning Support Fund (KIDSF) was established at the European Bank for Reconstruction and Development (EBRD) to finance decommissioning activities of Kozloduy 1-4 as well as support energy projects in the country. In October 2009 the European Commission granted Bulgaria an extra €300 million decommissioning, clean-up and waste treatment, bringing the overall compensation package to some €850 million by then[3]. At the end of 2013 a further €293 million was made available by the EU for units 1-4. EU support continues under the 2021-2027 multiannual financial framework.

In 2009 almost €120 million worth of contracts were signed under the KIDSF, including a €22.5 million contract for Onet Technologies for the treatment of radioactive concentrates and a €30 million contract for a joint venture between Iberdrola Ingenieria y Construccion and Belgoprocess NV to supply a waste treatment and conditioning facility.

Shutdown nuclear power reactors in Bulgaria

Reactor	Type	Model	Net MWe	First power	Shutdown
Kozloduy 1	VVER-440	V-230	405	7/74	12/02
Kozloduy 2	VVER-440	V-230	405	8/75	12/02
Kozloduy 3	VVER-440	Enhanced V-230	405	12/80	12/06
Kozloduy 4	VVER-440	Enhanced V-230	405	5/82	12/06

The licences for Kozloduy 1&2 were transferred to State Enterprise 'Radioactive Waste' (SE-RAW) by the Nuclear Regulatory Agency in October 2010, in anticipation of dismantling work. Decommissioning licences for them were issued by the NRA in 2014, and dismantling work along with management of materials and waste from decommissioning is to be conducted by a special division of State Enterprise Radioactive Waste Kozloduy (SE RAW Kozloduy). This will be paid for by the EBRD.

Research & development

The IRT–2000 pool-type research reactor was built by Moscow's Kurchatov Institute and went critical in 1959. Its original power was 1 MWt, but this was increased to 1.5 MWt in 1965 and to 2.0 MWt in 1970.

Highly enriched uranium (HEU) fuel (36% enriched) for the reactor was returned to Russia in 2003, and the HEU and LEU used fuel followed in 2008.[4] The reactor was shut down in 2008 to be rebuilt, but lack of funds has prevented completion of work.

Regulation, safety and non-proliferation

The Energy Ministry is responsible for the nuclear power industry. The two main entities are the Nuclear Regulatory Agency (NRA) set

up under the Safe Uses of Nuclear Energy Act 2002 and the Kozloduy Nuclear Power Plant plc. The NRA took over the functions of its predecessor, the Committee on the Safe Use of Atomic Energy for Peaceful Purposes (CUAEPP), originally set up under a 1985 Act.

The NRA is responsible for both regulation of nuclear installations in relation to safety and radiation protection, and also the management of radioactive wastes. It also undertook all nuclear functions related to European Union accession. The NRA has been a member of the Western European Nuclear Regulator's Association (WENRA) since 2003.

The Ministry of Health is also involved with radiation protection, and determines relevant standards. The Ministry of Environment and Water monitors radiation levels throughout the country.

Non-proliferation

Bulgaria is a party to the Nuclear Non-Proliferation Treaty (NPT) as a non-nuclear weapons state. Its safeguards agreement under the NPT came into force in 1972. It is member of the Nuclear Suppliers Group. In 2000 it signed the Additional Protocol in relation to its safeguards agreements with the International Atomic Energy Agency, and this is in force.

Notes & references

Notes

a. Over 90% of the country's gas supply comes from Russia. [Back]

b. The first four units (Kozloduy units 1-4) were all classed as VVER-440 V-230, even though the later pair (units 3&4) had many features of the later V-213 model. [Back]

c. The Belene plant was originally envisaged to be a VVER-1000 model V-320 plant. Several evolutionary versions of the V-320 exist, including the V-392 and the later V-466B, the model subsequently planned for Belene. The Belene plant type (AES-92, also referred to as the NPP-92) incorporates various reactor models – see also Reactor Technology section in Russian Nuclear Power page. [Back]

d. Negotiations on the early closure of Kozloduy 1-4 commenced in the early 1990s. In June 1993 the Nuclear Safety Account (NSA) Agreement was signed by the Bulgarian government, the Bulgarian National Electricity Company (NEK) and the European Bank for Reconstruction and Development (EBRD). Between 1991 and 1997, over 900 modifications to systems and equipment were carried out on Kozloduy units 1-4 in close consultation with the International Atomic Energy Agency (IAEA), the World Association of Nuclear Operators (WANO) and several other organizations to improve safety and bring them closer to international norms, especially for units 3&4. This short-term safety upgrading program cost some €130 million. Nevertheless, according to the NSA Agreement, Bulgaria had to close Kozloduy 1&2 in 1997 and units 3&4 in 1998. This agreement included €24 million for safety upgrades of the units before closure as well as measures to secure energy supply.

Following the failure of this agreement, in 1999 the European Commission (EC) and the government of Bulgaria signed a memorandum in which both sides agreed that units 1&2 must be closed before 2003 and units 3&4 before 2006 (according to the EC) or 2008 and 2010 respectively (according to the Bulgarian government). In return for early closure, Bulgaria was offered €200 million compensation, and this was increased to €550 million when 2006 was agreed as the shutdown dates for units 3&4. Over 1998-2002 a complex modernization programme approved by the Bulgarian Nuclear Regulatory Agency (NRA) was undertaken in line with IAEA safety criteria to bring the units into conformity with current world standards. This was only fully implemented on units 3&4. Units 1&2 were closed down at the end of 2002.

An IAEA mission in 2002 reported very favorably on units 3&4.[5] Summarizing the conclusions, the IAEA 2002 Annual Report states: "The team concluded that the operational and design safety at Kozloduy now corresponds to the level of improvements seen at plants of similar vintage elsewhere. Many of the safety measures adopted for these plants in the design, operation and seismic areas exceeded those that were foreseen"[6]. Later, in 2003 after a two-week scrutiny by 18 international inspectors, the World Association of Nuclear Operators (WANO) reported that units 3&4 met all necessary international standards for safe operation. Then, in November 2003 a review by 11 experts from the European Council's Atomic Questions Group (AQG) found that issues arising in the previous AQG review of 2001 had been addressed. In the meantime, the Bulgarian government tried to renegotiate the agreed 2006 shutdown for units 3&4 and gain a reprieve until the

licences expired (2011 and 2013), giving a 30-year operating lifetime.

The closure and subsequent decommissioning commitments concerning Kozloduy 1-4 are included in Article 30 of the *Act concerning the conditions of Accession of the Republic of Bulgaria and Romania and the adjustments to the treaties on which the European Union is founded* (2005 Act of Accession). Electricity shortages in the Balkan area became acute in 2007. Apparently units 3&4 could be brought back into operation in six months. Bulgaria's EU accession agreement says that in a national crisis the country has the right to resume power generation at Kozloduy 3&4. The EU is contributing to decommissioning costs for units 1-4.

See also information page on Early Soviet Reactors and EU Accession. [Back]

e. Kozloduy 3&4 had a number of important differences from the standard V-230 model that makes them closer to the more advanced V-213 design. These include a stainless steel-lined pressure vessel (alleviating vessel embrittlement), three safety system channels, and a low pressure emergency core cooling system. [Back]

f. Skoda Alliance proposed an upgraded V-320 model reactor based on the Temelin units in the Czech Republic, for €5 billion. (Skoda manufactured and delivered the V-320 reactor pressure vessel for Belene 1 in 1989.) The winning ASE bid proposed third-generation VVER-1000 reactors (model V-466), with the equipment supplied before 1991 being bought back, for use in Russia. The new units are based on the AES-91 plant built by ASE at Tianwan in China. NEK said that the AES-92 with a third-generation reactor satisfied stringent Western European safety standards and so would be more acceptable in the European Union, which Bulgaria joined in January 2007. [Back]

g. In 2007 NEK sought equity or strategic (customer) partners for a 49% interest in the project. Applications were received from ten companies; six were shortlisted in September and five subsequently bid in October 2007: Suez's Electrabel of Belgium, Enel of Italy, German companies RWE and E.ON, and CEZ of the Czech Republic (EDF dropped out of the bidding). In March 2008 NEK announced that RWE Power and Suez/ Electrabel were favoured, but binding offers for 49% equity were initially too low. However, in October 2008 NEK announced that RWE Power was to be the strategic investor, but intended to share the role. GDF Suez was initially interested, but declined in order to concentrate on other projects.

RWE was required to invest €1.275 billion as equity in the project, and provide a €300 million corporate loan in advance. Total cost of the new plant inflated to September 2009 then came to some €6 billion. Including owner's costs, infrastructure, grid uprating, site works, project management and finance it is likely to be about €10 billion.

In December 2008 the Belene Power Company (BPC) was set up as a joint venture between NEK (51%) and RWE Power (49%) to manage the work. Much of 2009 was to be taken up with BPC finalising project specifications to RWE's satisfaction. This would have been the first substantial Western investment in Russian nuclear technology since Finland's in the early 1970s.

Through 2009 following a change in government, concern was expressed as to whether the project was affordable. In September the government said that it would not take the full 51% stake in the plant and 20% looked more practical, to be held by Bulgarian Energy Holdings (BEH). RWE was also expected to reduce its stake, but in October pulled out altogether. [Back]

h. Early in 2010 a new project company financed by Russia was established to finance construction activities until a strategic investor is found. The funds would be repaid by a future investor, or some could remain invested on a permanent basis through a share in Belene's equity capital. Director General of Rosatom, Sergey Kiriyenko, said: "In our opinion, the Belene NPP project is viable therefore we are ready to finance it temporarily until a new strategic investor is found." According to Kiriyenko, €400-450 million was needed up to the end of 2010, and up to €1.5 billion more in 2011.[7] However, while the deal was agreed in principle[8], the terms were not finalized and in April 2010 the government declined the offered Russian financing and said it would look for a European partner[9]. [Back]

i. There was considerable uncertainty over the cost of the Belene project. Though the 18 January 2008 contract states a price of €3.997 billion, the relevant parties disagree over how this price should be adjusted for inflation[10]. At the end of September 2010 it was reported that the prime minister said: "I would like to reduce the price of the nuclear plant by €700-800 million, down from over €7 billion."[11] However, although the November 2010 memorandum of understanding (MoU) itself does not provide for an overall price, several reports on the MoU quote a maximum fixed price by 2016 of €6.298 billion[12]. [Back]

j. Up to 1989 used fuel was sent to the Mayak facility in Russia for reprocessing. A new agreement on reprocessing used fuel was reached in 1998. This agreement is conditional upon the future return of the high-level waste from reprocessing. [Back]

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